

Félix Gélinas

Toronto, Ontario, Canada

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Education

York University (Advisors: Nantel Bergeron and Vincent Pilaud) <i>Doctor of Philosophy in Mathematics (PhD)</i>	2023 – expected 2027 Toronto, Ontario, Canada
University of Toronto (Advisor: Swastik Kopparty) <i>Master of Science in Mathematics (MSc)</i>	2022 – 2023 Toronto, Ontario, Canada
Université du Québec à Montréal (UQAM) <i>Bachelor of Science in pure mathematics (BSc)</i>	2019 – 2022 Montréal, Québec, Canada

Research Experience

PhD project (Advisors: Nantel Bergeron and Vincent Pilaud) <i>York University</i>	September, 2023 – Present Toronto, Ontario, Canada
Supervision of Summer internship in Research Groups (York) <i>York University</i>	May, 2024 – August, 2024 Toronto, Ontario, Canada
Group Working On Hopf Algebra • Working with Mike Zabrocki, Lucas Gagnon • Supervising students Eric Schlums and Nicolas Andrews • Writing of an article • Programming (SageMath) • Hopf Algebra	
Group Working On Machine Learning • Working with Mike Zabrocki, Lucas Gagnon • Supervising students Eniola Ogunjimi and Inderpreet Ganda • Writing of an article • Programming (SageMath) • Machine Learning, Combinatorics	
Master Thesis project (Advisor: Swastik Kopparty (UofT)) <i>University of Toronto (UofT)</i>	May, 2023 – August, 2023 Toronto, Ontario, Canada
• Summarizing articles • Writing of a rapport of articles • Reading articles • Combinatorics, Graph Theory	
Summer internship in Research (Advisor: Hugh Thomas (UQAM)) <i>Université du Québec à Montréal (UQAM)</i>	May, 2022 – August, 2022 Montréal, Québec, Canada
• Writing of an article • Programming (SageMath) • Combinatorics, Dyck path	
Summer internship in Research (Advisor: Hugh Thomas (UQAM)) <i>Université du Québec à Montréal (UQAM)</i>	May, 2021 – August, 2021 Montréal, Québec, Canada
• Writing of an article • Programming (SageMath) • Algebra, Group theory	

Teaching Assistant (TA)

Teaching Assistant (TA)

September 2025 – Present

York University

Toronto, Ontario, Canada

- Applied Calculus I (MATH1013)
- Discrete Mathematics for Computer Science (MATH1019)

Teaching Assistant (TA)

May 2025 – August 2025

York University

Toronto, Ontario, Canada

- Introduction to Logic for Computer Science (MATH1090)

Teaching Assistant (TA)

January 2025 – April 2025

York University

Toronto, Ontario, Canada

- Introduction to Logic for Computer Science (MATH1090)
- Problems, Conjectures and Proofs (MATH1200)

Teaching Assistant (TA)

September 2024 – December 2024

York University

Toronto, Ontario, Canada

- Differential Calculus with Applications (MATH1300)
- Introduction to Logic for Computer Science (MATH1090)
- Problems, Conjectures and Proofs (MATH1200)

Teaching Assistant (TA)

May 2024 – June 2024

York University

Toronto, Ontario, Canada

- Applied Multivariate and Vector Calculus (MATH2015)

Teaching Assistant (TA)

January 2024 – April 2024

York University

Toronto, Ontario, Canada

- Integral Calculus with Applications (MATH1310)
- Problems, Conjectures and Proofs (MATH1200)

Teaching Assistant (TA)

September 2023, December 2023

York University

Toronto, Ontario, Canada

- Problems, Conjectures and Proofs (MATH1200)
- Differential Calculus with Applications (MATH1300)

Teaching Assistant (TA)

September, 2022 – April 2023

University of Toronto

Toronto, Ontario, Canada

- Calculus I (MAT135)
- Calculus II (MAT136)

Teaching Assistant (TA)

January, 2022 – April 2022

Université du Québec à Montréal (UQAM)

Montréal, Québec, Canada

- Calculus of Differential Forms (MAT2410)

Organization

Co-organizer for Participants-Seminar of the CGGC

October, 2025 – November 2025

Universitat de Barcelona (CRM)

Barcelona, Spain

- Co-organizer for Participants-Seminar of the CGGC Research School (Weekly from October 23 to November 20, 2025)

Other Experience

Moderator for Mathematical Summer in Paris

July, 2021 – July 2021

Université Paris-Saclay

Paris, France

- Moderator for an online mathematical conference (from July 5 to 9, 2021)

Preprint

Source Characterization of the Hypergraphic Posets. (<https://arxiv.org/abs/2508.16006>) – Félix Gélinas.

Ornamentation Lattices and Intreeval Hypergraphic Lattices. (<https://arxiv.org/abs/2508.01606>) – Antoine Abram, Jose Bastidas, Félix Gélinas, Vincent Pilaud, Andrew Sack.

When are Hopf Algebras Determined by Integer Sequences? (<https://arxiv.org/abs/2505.06941>) – Nicolas Andrews, Lucas Gagnon, Félix Gélinas, Eric Schlums, Mike Zabrocki.

Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders (<https://arxiv.org/abs/2212.12171>) – Félix Gélinas, Adrien Segovia, Hugh Thomas. Accepted in The Electronic Journal of Combinatorics.

Publications

Gabriel Frieden, Félix Gélinas, and Étienne Soucy. “Cones and ping-pong in three dimensions”. In: *Involve, a Journal of Mathematics* 17.1 (Mar. 2024), pp. 11–28. ISSN: 1944-4176. DOI: 10.2140/involve.2024.17.11. URL: <http://dx.doi.org/10.2140/involve.2024.17.11>

Presentations (Talks)

Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM) May, 2025
University of Ottawa Ottawa, Ontario, Canada

- Source Characterization of the Hypergraphic Posets
- Conference Program

Algebraic and enumerative combinatorics seminar May, 2025
University of Waterloo Waterloo, Ontario, Canada

- Source Characterization of the Hypergraphic Posets
- Seminar Program

Winter School and Workshop on Polytopes February, 2025
Universität Osnabrück Osnabrück, Lower Saxony, Germany

- Source Characterization of the Hypergraphic Posets
- Conference Program

Séminaire du LACIM February, 2025
Université du Québec à Montréal (UQAM) Montréal, Québec, Canada

- Caractérisation du poset hypergraphique à partir des sources d'orientations acycliques
- Seminar Program

Combinatorial Algebra meets Algebraic Combinatorics (CAAC) January, 2025
The Fields Institute Toronto, Ontario, Canada

- Source Characterization of the Hypergraphic Posets
- CAAC Program

The Algebraic Combinatorics Workgroup October-November, 2024
Nantel Bergeron (York University) – Vasu Tewari (UofT) The Fields Institute, Toronto, Ontario, Canada

- Hypergraphic polytopes | Part 1 - Part 2
- Part 1 : www.fields.utoronto.ca/talks
- Part 2 : www.fields.utoronto.ca/talks

The Algebraic Combinatorics Workgroup

May, 2024

*Nantel Bergeron– Kelvin Chan– Blessing Oni (York University)**The Fields Institute, Toronto, Ontario, Canada*

- Gently Discovering Gentle Algebras
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

Graduate student Group

May, 2024

*Leo Jiang (UofT)**The Fields Institute, Toronto, Ontario, Canada*

- Permutrees

The Algebraic Combinatorics Workgroup

September, 2023

*Nantel Bergeron– Kelvin Chan– Blessing Oni (York University)**The Fields Institute, Toronto, Ontario, Canada*

- The volume polynomial of a Matroid
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

The Graduate Student Seminar UofT

March, 2023

*Mathematics Departement UofT**Toronto, Ontario, Canada*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders

The Algebraic Combinatorics Workgroup

March, 2023

*Nantel Bergeron (York University)– Kelvin Chan (York University)**The Fields Institute, Toronto, Ontario, Canada*

- Cones and ping-pong in three dimensions
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

Applied Algebra Seminar

March, 2023

*Lucas Gagnon (York University)**York, Ontario, Canada*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders
- <https://garsia.math.yorku.ca/seminar/algebra.html>

The Graduate Students Seminar (UQAM)

August, 2022

*Université du Québec à Montréal (UQAM)**Montréal, Québec, Canada*

- Cones and ping-pong in three dimensions

Pan-Québec colloquium for students of the ISM

June, 2022

*Institute of Mathematical Sciences (ISM)**Québec, Québec, Canada*

- Cones and ping-pong in three dimensions
- <http://ism.uqam.ca/Colloque-panquebecois>

*Presentations (Poster)***AlCoVe : an Algebraic Combinatorics Virtual Expedition**

May, 2025

Online Conference

- Source Characterization of the Hypergraphic Posets
- AlCoVe Program

VIII Encuentro Colombiano de Combinatoria (ECCO)

June, 2024

*CIMPA Research School**Popayán, Cauca, Colombia*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders
- ECCO Program

Combinatorial Algebra meets Algebraic Combinatorics (CAAC)

January, 2024

*Université du Québec à Montréal**Montréal, Québec, Canada*

- The volume polynomial of a Matroid
- CAAC Program

Combinatorial Algebra meets Algebraic Combinatorics (CAAC)

January, 2023

*University of Waterloo**Waterloo, Ontario, Canada*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders
- <https://sites.google.com/view/caac2023/schedule>
- Link to the poster

Awards & Honors

FGS Tuition Fee Bursary	2025
<i>York University</i>	
Kingsley Abu Imonikhe Graduate Award	2025
<i>York University</i>	
Academic Excellence Fund	2025
<i>York University</i>	
Vernon Oliver Stong Graduate Scholarships in Science	2024
<i>York University</i>	
Graduate Studies	2023-2027
<i>York University</i>	
Graduate Studies	2023-2024
<i>York University</i>	
Graduate Studies	2022-2023
<i>University of Toronto</i>	
Undergraduate Student Research Awards	2022
<i>NSERC</i>	
Advisor contribution NSERC	2022
<i>Hugh Thomas</i>	
Supplements of the NSERC Undergraduate Student Research Awards	2022
<i>FRQNT</i>	
Undergraduate summer scholarship	2021
<i>Institute of Mathematical Sciences (ISM)</i>	
Advisor contribution ISM	2021
<i>Hugh Thomas</i>	
Advisor contribution ISM	2021
<i>Christophe Reutenauer</i>	

Summer Schools

VIII Encuentro Colombiano de Combinatoria (ECCO)	2024
<i>CIMPA Research School, Popayán, Cauca, Colombia</i>	
ISM Discovery School: Advances in Geometric Combinatorics and Representation Theory	2024
<i>R. González D'León – A. H. Morales – F. Saliola – H. Thomas, Montréal, Québec, Canada</i>	

Winter Schools

Winter School and Workshop on Polytopes	2025
<i>M. Juhnke – M. Reitzner – C. Thäle, Osnabrück, Lower Saxony, Germany</i>	

Intensive Research Program / Workshop

Combinatorial Geometries and Geometric Combinatorics	2025
<i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	
Lattice Theory Workshop	2025
<i>The Banff International Research Station (BIRS), Banff, Alberta, Canada</i>	

Seminars & Conferences attended

Recollections on polyhedral geometry and (oriented) matroids <i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	2025
Seminar on Pure Mathematics <i>The Hong Kong University of Science & Technology, New Territories, Sai Kung, Hong Kong</i>	2025
AlCoVe : an Algebraic Combinatorics Virtual Expedition <i>L. Colmenarejo – M. Gillespie – O. Pechenik – L. Solus – F. Tom – L. Vecchi, Online Conference</i>	2025
Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM) <i>Ottawa, Ontario, Canada</i>	2025
Algebraic and Enumerative Combinatorics Seminar of the University of Waterloo <i>Karen Yeats, Waterloo, Ontario, Canada</i>	2025
Combinatorics Seminar of UofT <i>Vasu Tewari, Toronto, Ontario, Canada</i>	2025
Winter School and Workshop on Polytopes <i>M. Juhnke – M. Reitzner Osnabrück – C. Thäle, Lower Saxony, Germany</i>	2025
LACIM seminar <i>J. Bastidas – S. Giraudo – C. Reutenauer, Montréal, Québec, Canada</i>	2025
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) <i>S. Faridi – S. Kara – R. Orellana – G. Smith – V. Tewari – M. Zabrocki, Toronto, Ontario, Canada</i>	2025
The Algebraic Combinatorics Workgroup <i>Nantel Bergeron – Kelvin Chan – Vasu Tewari, The Fields Institute, Toronto, Ontario, Canada</i>	2022-2025
Applied Algebra Seminar <i>Lucas Gagnon, York, Ontario, Canada</i>	2022-2025
Workshop on Computational and Applied Enumerative Geometry <i>T. Brysiewicz – L. Garcia Puente – N. Hein – J. Lopez Garcia – K. Purbhoo, Toronto, Ontario, Canada</i>	2024
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) <i>S. Brauner – J. Levinson – A. Morales – G. Park – F. Saliola – H. Thomas, Montréal, Québec, Canada</i>	2024
The Graduate Student Seminar UofT <i>Graduate Student of UofT, Toronto, Ontario, Canada</i>	2023
Matroids - Combinatorics, Algebra and Geometry Seminar <i>Ahmed Ashraf, The Fields Institute, Toronto, Ontario, Canada</i>	2023
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) <i>Sara Faridi–Selvi Kara –Oliver Pechenik –Matthew Satriano, Waterloo, Ontario, Canada</i>	2023
The Graduate Students Seminar (UQAM) <i>Université du Québec à Montréal (UQAM), Montréal, Québec, Canada</i>	2022
Non-Commutative Algebra, Representation Theory, Special Functions (Conference) <i>CRM, Montréal, Québec, Canada</i>	2022
Pan-Quebec colloquium for students of the ISM <i>Institute of Mathematical Sciences (ISM), Université Laval, Québec, Québec, Canada</i>	2022
Open Problems in Algebraic Combinatorics (OPAC) <i>University of Minnesota, Minneapolis, Minnesota, USA</i>	2022
Graduate Student Conference on Combinatorics, Commutative Algebra and Geometry <i>University of Minnesota, Minneapolis, Minnesota, USA</i>	2022
LACIM seminar <i>LACIM, Montréal, Québec, Canada</i>	2021-2022

Interests

Combinatorics

Partially Ordered Set (Poset)

Algebraic Combinatorics

Enumerative Combinatorics

Graph Theory

Representation Theory

Algebra

Commutative Algebra

Algebraic Topology